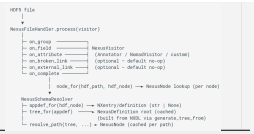
TASK DESCRIPTION

The task consists of systematically analysing the pynxtools ecosystem to identify and document the individual readers, parsers, and format handlers implemented within each plugin.

For each handler, the supported input formats, data-processing procedures, unit conversions, mathematical transformations, metadata-mapping rules, and assumptions regarding data structure and interpretation are examined.

The implemented behaviour is compared against the relevant NeXus application definitions, data-format specifications, and documented conventions to identify non-standard processing, implementation-specific deviations, and implicit assumptions. Each finding is documented in a structured table containing the plugin and handler names, source-code repository, relevant documentation and implementation locations, a description of the identified behaviour, and supporting evidence from the source code, tests, or technical documentation.

The resulting inventory provides a traceable overview of format-specific processing and facilitates the assessment of consistency, interoperability, and compliance with the intended NeXus data model.





GENERAL DOCUMENTATION:

https://fairmat-nfdi.github.io/pynxtools/tutorial/build-a-reader.html
https://fairmat-nfdi.github.io/pynxtools/how-tos/pynxtools/build-a-plugin.html
https://fairmat-nfdi.github.io/pynxtools/how-tos/pynxtools/using-pynxtools-test-framework.html
https://fairmat-nfdi.github.io/pynxtools/how-tos/pynxtools/use-multi-format-reader.html
https://fairmat-nfdi.github.io/pynxtools/how-tos/pynxtools/use-with-mymrd.html
https://fairmat-nfdi.github.io/pynxtools/learn/pynxtools/overview.html

Plugin	State	Field	(Relevant) NeXus Application Definition	URL	Documentation	Reader	Known issues
pynxtools-apm		Atom probe microscopy and field ion microscopy	NXapm	GitHub: pynxtools_apm	Docs: pynxtools-apm	Reader: pynxtools-apm	Known issues: pynxtools-apm
pynxtools-camels	Public_under_dev	Plugin containing NOMAD search app for automation project (NOMAD CAMELS search app for NeXus CAMELS entries)	-	GitHub: pynxtools-camels	-	[MISSING]	-
pynxtools-ellips	Public	Ellipsometry	NXellipsometry	GitHub: pynxtools-ellips	Docs: pynxtools-ellips [INCOMPLETE]	Reader: pynxtools-ellips	-
pynxtools-em	Public	Electron microscopy	NXem	GitHub: pynxtools-em	Docs: pynxtools-em	Reader: pynxtools-em	-
pynxtools-em-insitu	Private	Electron microscopy demo	?	GitHub: pynxtools-em-insitu	-	Reader: pynxtools-em-insitu	-
pynxtools-focus	Private	Focus tools [dermotdev state]	?	GitHub: pynxtools-focus	-	Reader: pynxtools-focus	-
pynxtools-igor	Public	Wavemetrics Igor Pro	No single technique-specific definition	GitHub: pynxtools-igor	Docs: pynxtools-igor [INCOMPLETE]	Reader: pynxtools-igor	-
pynxtools-mpes	Public	Multidimensional photoelectron spectroscopy	NXmpes NXmpes_arpes	GitHub: pynxtools-mpes	Docs: pynxtools-mpes	Reader: pynxtools-mpes	-
pynxtools-microstructure	Public	Microstructure	NXmicrostructure	GitHub: pynxtools-microstructure	Docs: pynxtools-microstructure [no reference]	Reader: pynxtools-microstructure	-
pynxtools-raman	Public	Raman	NXraman	GitHub: pynxtools-raman	Docs: pynxtools-raman	Reader: pynxtools-raman	-
pynxtools-xpm	Public	Scanning probe microscopy	NXspm NXafm NXstm	GitHub: pynxtools-xpm	Docs: pynxtools-xpm	Reader: pynxtools-xpm	-
pynxtools-stm	Public archive	Scanning tunneling microscopy and scanning tunneling spectroscopy	NXstm NXem	GitHub: pynxtools-stm	Docs: pynxtools-stm	Reader: pynxtools-stm	-
pynxtools-xas	Public	X-ray absorption spectroscopy (XAS)	NXxas	GitHub: pynxtools-xas	Docs: pynxtools-xas [Learn section missing]	Reader: pynxtools-xas	
pynxtools-xps	Public	X-ray photoelectron spectroscopy	NXxps	GitHub: pynxtools-xps	Docs: pynxtools-xps	Reader: pynxtools-xps	-
pynxtools-xrd	Public	X-ray diffraction	NXxrd NXxrd_pan	GitHub: pynxtools-xrd	Docs: pynxtools-xrd	Reader: pynxtools-xrd	-